

Frontend: Theory Assignment 2

Q1. What is CSS Box Model?

Every element in a webpage is treated like a box. This box model tells how the size of the box is made up. It has 4 parts - content, padding, border and margin.

Content is the actual text or image. Padding is the space between content and border. Border is the line around the padding. Margin is the space outside the border, between this element and the next element.

Q2. What is media query? Explain with a sample code.

A media query is used in CSS to apply different styles based on the screen size. It is mainly used to make a website responsive so it looks fine on mobile, tablet and desktop.

```
body {  
  background-color: lightblue;  
}  
  
@media (max-width: 600px) {  
  body {  
    background-color: lightgreen;  
  }  
}
```

Here, when the screen width becomes 600px or less, the background color changes to light green.

Q3. What are the features of Bootstrap?

- It is a free and open source CSS framework.
- It has a responsive grid system.
- It gives ready made components like buttons, navbars, cards and forms.
- The website works well on mobile, tablet and desktop.
- It has built in JavaScript plugins for modals, carousels etc.
- It saves time as we don't have to write CSS from scratch.

Q4. Write an example of basic grid structure in Bootstrap?

Bootstrap grid is based on a 12 column layout. We use container, row and col classes together.

```
<div class="container">  
  <div class="row">  
    <div class="col">Column 1</div>  
    <div class="col">Column 2</div>  
    <div class="col">Column 3</div>  
  </div>  
</div>
```

Q5. What are CSS Selectors?

CSS selectors are used to select the HTML element we want to style.

- Element selector - selects all elements of a tag. Example: p { }
- Class selector - selects elements with a class. Example: .box { }
- ID selector - selects one element with an id. Example: #header { }
- Universal selector - selects all elements. Example: * { }
- Group selector - applies same style to many selectors. Example: h1, h2 { }

Q6. What are CSS backgrounds, list the properties?

CSS background properties are used to set the background of an element.

- background-color - sets background color.
- background-image - sets an image as background.
- background-repeat - decides if the image repeats or not.
- background-position - sets position of background image.
- background-size - sets size of background image.
- background-attachment - decides if background stays fixed or scrolls.

Q7. How can we overlap elements in CSS?

We can overlap elements using position property with z-index. First give the elements position: relative or absolute, then use z-index to decide which element shows on top.

Higher z-index value is shown above lower z-index value.

Q8. What are the different positioning properties in CSS?

- static - default position, element is in normal flow.
- relative - positioned relative to its own normal position.
- absolute - positioned relative to its nearest positioned parent.
- fixed - stays at same place even when page is scrolled.
- sticky - acts like relative first, then becomes fixed after a scroll point.

Q9. What is ECMAScript?

ECMAScript is the standard on which JavaScript is based. It defines the rules and features a scripting language should follow, like syntax, data types and objects. Different versions like ES5, ES6 added new features to JavaScript, for example let, const and arrow functions came in ES6.

Q10. What is the difference between var, let and const in JavaScript. Explain Temporal Dead Zone?

- var - function scoped, can be re-declared and updated, hoisted with value undefined.
- let - block scoped, can be updated but not re-declared in same scope, hoisted but not initialized.
- const - block scoped, cannot be updated or re-declared, must be given value at declaration.

Temporal Dead Zone is the time between a let or const variable getting hoisted and it actually getting declared in the code. If we try to use the variable in this time, we get an error. This does not happen with var since it is set to undefined during hoisting.